

Specifications

Sensor Specifications

- **Spectral Band:** LWIR (8 μm - 12 μm)
- **Uncooled Detector**
- **Pixel Size:** $p = 12 \mu\text{m}$
- **Dimensions:** (768 x 1024) pixels
- **Window:** 0.62 mm thickness

Optical System without Phase Mask

Refer to the Technical Details section below for more precise values and constraints

Target Value	Allowed Margin / Constraint	Comments
Focal Length	14 mm	5% => [13.86 ; 14.14] mm
Aperture Number (N)	Minimum $N < 2$	The aperture must be as large as possible.
Total Field of View	6.144 mm	2/3 of the total field => 4.1 mm acceptable
MTF (Modulation Transfer Function)	Diffraction Limit	20% deviation from the diffraction limit for 2/3 of the total field at Nyquist frequency (f_c)
Operating Temperature	X	Operational between -20°C and 90°C
Lens Thickness	X	$> \Phi/10$
Window-to-Sensor Distance	0.35 mm	[0.1 ; 0.6] mm
Total System Size	20 mm in length, 20 mm in height	50% => 30 mm

Technical Details:

- Nyquist Frequency $f_c = 1/2p = 41.6 \text{ mm}^{-1}$
- Using the Pythagorean theorem:
 $d = \sqrt{[(1024 \cdot p/2)^2 + (768 \cdot p/2)^2]} = 6,144 \text{ mm}$

